

| S — Set (session) |             |                                                                                                          |
|-------------------|-------------|----------------------------------------------------------------------------------------------------------|
| Short             | Long        | Description                                                                                              |
| SS                | SetSpeed    | Set cruise speed mm/s ( $\leq$ max_speed); bare reloads init_speed; applies live to the next fill.       |
| SA                | SetAccel    | Set accel mm/s <sup>2</sup> ( $\leq$ max_accel); bare reloads init_accel; applies live to the next fill. |
| SE                | SetEnable   | Driver enable 0 1; bare toggles; required before motion; off stops hard.                                 |
| ST                | SetTerminal | Terminal Mode 0 1; bare toggles; local echo + UART command sniff to USB.                                 |
| SV                | SetVerbose  | Verbose status push 0 1; bare toggles; ~3 Hz #M/#I lines when on.                                        |
| SD                | SetDebug    | USB-only debug level 0..5; bare restores default; never sent on UIC UART.                                |

| G — Get (session) |             |                                                               |
|-------------------|-------------|---------------------------------------------------------------|
| Short             | Long        | Description                                                   |
| GS                | GetSpeed    | Reply GS:<mm/s> — current session cruise speed.               |
| GA                | GetAccel    | Reply GA:<mm/s <sup>2</sup> > — current session acceleration. |
| GE                | GetEnable   | Reply GE:0 1 — driver enable state.                           |
| GT                | GetTerminal | Reply GT:0 1 — Terminal Mode state.                           |
| GV                | GetVerbose  | Reply GV:0 1 — verbose push state.                            |
| GD                | GetDebug    | Reply GD:<0..5> — USB debug level.                            |

| I — Is / status |            |                                                                        |
|-----------------|------------|------------------------------------------------------------------------|
| Short           | Long       | Description                                                            |
| IM              | IsMoving   | Reply IM:0 1 — axis currently moving (or settling).                    |
| IH              | IsHoming   | Reply IH:0 1 — homing cycle active.                                    |
| IL              | IsLimit    | Reply IL:0 1 — at soft-limit position.                                 |
| IE              | IsError    | Reply IE:0 1 — PIN_DRV_ERROR / EMO latched.                            |
| IP              | IsPosition | Reply IP:<mm> — current planner position.                              |
| IT              | IsTarget   | Reply IT:<mm> - — seek target, or - if none.                           |
| IR              | IsReady    | Reply IR:1 only if idle, not homing, enabled, and not waiting.         |
| IW              | IsWaiting  | Reply IW:1 if any W / WM / WH wait is active.                          |
| ID              | IsDiag     | Reply underrun count, peak STEP Hz, overshoot steps, min FIFO level.   |
| IZ              | IsReset    | Reply last chip reset cause (power wdt run soft debug brownout ...).   |
| IX              | Pinout     | ASCII table: GP number, pin name, brief description ( $\leq$ 80 cols). |

| M — Movement |           |                                                                     |
|--------------|-----------|---------------------------------------------------------------------|
| Short        | Long      | Description                                                         |
| MT           | MoveTo    | Move to absolute mm; needs enable; live-retargets an active move.   |
| M            | Move      | Relative move by mm (alias MoveBy); needs enable; live-retargets.   |
| ML           | MoveLeft  | Continuous jog negative; soft-stop with MS or !.                    |
| MR           | MoveRight | Continuous jog positive; soft-stop with MS or !.                    |
| MH           | MoveHome  | Homing cycle (home_mode); no-op if 0; needs SE 1; cancel with MS/H. |
| MS           | MoveStop  | Soft decelerate to stop; keeps enable; does not cancel waits.       |

| X — Extender |        |                                                                     |
|--------------|--------|---------------------------------------------------------------------|
| Short        | Long   | Description                                                         |
| X0~9         | Ext0~9 | Ext out n logical 0 1; bare toggles; glued X00≡X0 0; ok during EMO. |

| C — Config |           |                                                                                         |
|------------|-----------|-----------------------------------------------------------------------------------------|
| Short      | Long      | Description                                                                             |
| CS         | ConfigSet | Set persistent key value (mc.ini); silent ok; updates session init_speed/init_accel/... |
| CG         | ConfigGet | Get key → CG:key=value; bare dumps all keys.                                            |

| W — Wait |            |                                                                          |
|----------|------------|--------------------------------------------------------------------------|
| Short    | Long       | Description                                                              |
| W        | Wait       | Delay sec then continue ; chain; bare → 1 s; never !E:timeout.           |
| WM       | WaitMoving | Pause chain until move ends; optional timeout cancels remaining chain.   |
| WH       | WaitHoming | Pause chain until homing ends; optional timeout cancels remaining chain. |

| V — Version |                 |                                                 |
|-------------|-----------------|-------------------------------------------------|
| Short       | Long            | Description                                     |
| VA          | VersionAbout    | Reply VA: about string (name, version, author). |
| VF          | VersionFW       | Reply VF:<version> — firmware version.          |
| VP          | VersionProtocol | Reply VP:<n> — protocol version.                |

| Special |      |                                                                       |
|---------|------|-----------------------------------------------------------------------|
| Short   | Long | Description                                                           |
| H/HT    | Halt | Immediate STEP abort; enable off; cancel waits and remaining ; chain. |
| P       | Pins | List PIN_*=GPIO lines (machine-readable, read-only).                  |
| \$/HL   | Help | ASCII table of all commands ( $\leq$ 80 columns).                     |

**Wire:** one command per line (`\n`); chain with `;`; `#` comment to EOL (comment-only lines ignored); bare bool setters toggle; motion/settings silent on success; errors `!E:code message`.  
**Realtime** (no newline): `? status · ! soft stop · Ctrl-X (0x18) soft reset`.  
**Status** (`#X ...`): `I` idle · `A` accel · `M` cruise · `B` decel · `H` homing · `L` hard-limit · `D` disabled · `E` error. Moving: `#M/#A/#B pos speed accel [target]`.  
**Halt vs Stop:** `MS/!` soft decel (enable kept); `H/HT` immediate abort, enable off, cancel waits.